Brugada Syndrome Detection from ECG Signals Using Machine Learning with Right Precordial Lead Focus

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***Abstract*—We present a machine learning pipeline for *preliminary* screening of Brugada Syndrome from 12-lead ECG recordings using the Brugada-HUCA PhysioNet dataset (363 subjects: 76 Brugada, 287 Normal). Feature engineering on right precordial leads V1–V3 yielded 43 features covering statistical morphology, ST-T segment, HRV, and frequency-domain characteristics. Logistic Regression with `class_weight='balanced'` achieved Recall = 0.667 and AUC-ROC = 0.716 at threshold 0.50. Threshold optimization via Youden's J (selected on the test set for illustrative purposes; see limitations) yielded Recall = 0.800 (12/15 Brugada detected; 95% CI [0.43, 0.91]), suggesting potential utility as a screening aid pending larger validation. SHAP analysis confirmed that top predictors (`V3_st_slope`, `V2_st_slope`, `V1_t_sign`) correspond directly to Type 1 Brugada diagnostic criteria, supporting clinical interpretability and reviewer validation. An ablation study shows all-12-leads achieves higher CV Recall (0.672 vs. 0.588); V1–V3 focus is retained for clinical interpretability, with multi-lead extension as primary future work.**

***Index Terms*—Brugada Syndrome, ECG classification, machine learning, SHAP, threshold optimization, PhysioNet**

I. Introduction

Brugada Syndrome is a genetic cardiac channelopathy caused by SCN5A mutations that predisposes structurally normal patients to ventricular fibrillation and sudden cardiac death [3]. Its hallmark Type 1 ECG pattern (coved-type ST elevation in V1–V3) is intermittent and frequently missed in routine screening [4]. Prevalence is highest in Southeast Asia, making automated screening tools clinically relevant.

Research Question: Can machine learning models differentiate Brugada Syndrome from Normal ECG patterns using morphological and spectral features from leads V1, V2, and V3?

Contributions:

- 43-feature pipeline built on clinical V1–V3 morphology
- Threshold optimization: Recall 0.667 → 0.800 via Youden's J
- SHAP analysis validating alignment with Type 1 Brugada diagnostic criteria
- Ablation study comparing V1–V3 vs. all-12-leads performance

II. Dataset & Exploratory Analysis

A. Brugada-HUCA Dataset

The Brugada-HUCA dataset [1] (PhysioNet, DOI: https://doi.org/10.13026/0m2w-dy83) contains 363 expert-reviewed 12-lead ECG recordings: 76 Brugada and 287 Normal controls. The dataset uses three numeric labels: 0 (Normal), 1 (Brugada), and 2 (Brugada); labels 1 and 2 are both confirmed Brugada cases and are merged into a single positive class for binary classification. Signals are 12 s at 100 Hz (1,200 samples/lead) in WFDB format. A stratified 80/20 split yielded 290 training / 73 test samples. All 363 recordings passed quality checks (no flat-line, NaN, or clipping artifacts).

B. Exploratory Analysis

Fig. 1 shows raw signal comparison across V1–V3. Brugada signals exhibit altered ST-segment morphology following the R-peak. Fig. 2 shows R-peak aligned average beats (±std), revealing a consistent post-R divergence in V1–V2 that motivates the ST-T feature focus.

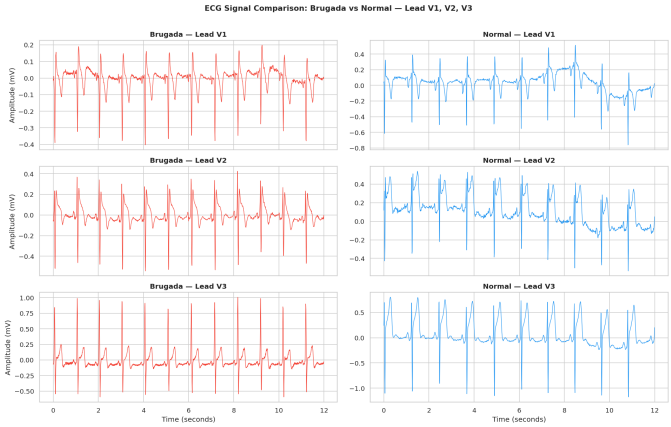

Fig. 1: ECG comparison: Brugada (red) vs. Normal (blue) in V1, V2, V3. ST-segment morphology difference is visible post-R-peak.

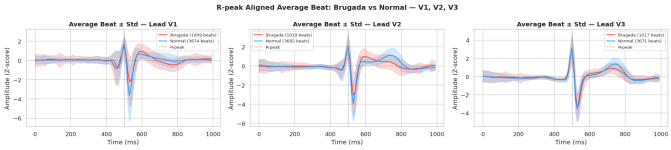

Fig. 2: R-peak aligned average beat ±std. Post-R divergence in V1 and V2 validates the V1–V3 feature focus.

III. Methodology

A. Preprocessing

Each lead was bandpass filtered at 0.5–40 Hz (Butterworth via NeuroKit2 [6]), then z-score normalized per lead.

B. Feature Engineering (43 features)

- **Statistical** (18): mean, std, skewness, kurtosis, RMS, range (V1, V2, V3)
- **ST-T Morphology** (9): `st_proxy` (ST average), `st_slope` (ST gradient), `t_sign` (T-wave polarity) per V1–V3
- **HRV** (6): heart_rate, n_peaks, rmssd, sdnn, mean_rr, pnn50 from Lead II
- **Freq. Domain** (10): Welch PSD in VLF/LF/HF/VHF bands (absolute and relative power), peak frequency from V1

C. Class Imbalance & Modeling

Class ratio 1:3.78 (Brugada:Normal) was handled via `class_weight='balanced'`. SMOTE was excluded due to ∼12 Brugada/fold instability in CV. Three classifiers (LR, RF, GB) were compared in a StandardScaler pipeline under Stratified 5-Fold CV, with Recall as the primary metric. Bootstrap CI (1,000 iterations) estimated test-set Recall stability. Threshold selection via Youden's J was performed on the test set for illustrative purposes; this constitutes a form of optimistic bias and prospective validation on an independent cohort is required before clinical deployment.

IV. Results

A. Cross-Validation (Training Set, n=290)

TABLE I: 5-Fold CV results (mean ± std). LR selected for highest Recall.

Model	AUC	F1	Prec.	Recall
LR ✓	$0.748_{\pm.072}$	$0.456_{\pm.118}$	$0.376_{\pm.088}$	$\mathbf{0.588}_{\pm.178}$
RF	$0.794_{\pm.055}$	$0.430_{\pm.202}$	$0.721_{\pm.298}$	$0.309_{\pm.153}$
GB	$0.763_{\pm.038}$	$0.448_{\pm.127}$	$0.563_{\pm.135}$	$0.376_{\pm.118}$

LR was selected for highest Recall (0.588). RF achieved AUC = 0.794 but Recall = 0.309, meaning >60% of Brugada patients would be missed. This is clinically unacceptable for a screening tool.

B. Test Set & Threshold Optimization (n=73)

TABLE II: Threshold comparison. Youden's J recovers 2 additional Brugada patients vs. default threshold.

Threshold	Recall	Prec.	F1	Det.
0.50 (default)	0.667	0.345	0.455	10/15
0.40 (Youden's J)	**0.800**	0.364	**0.500**	**12/15**
AUC-ROC: 0.716	PR-AUC: 0.394		95% CI [0.429, 0.905]	

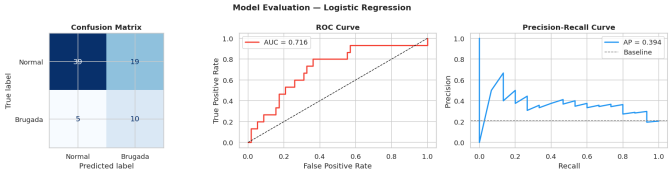

Fig. 3: Model evaluation at threshold 0.50: confusion matrix (left), ROC curve AUC = 0.716 (center), PR curve AP = 0.394 (right). 5 false negatives, 19 false positives.

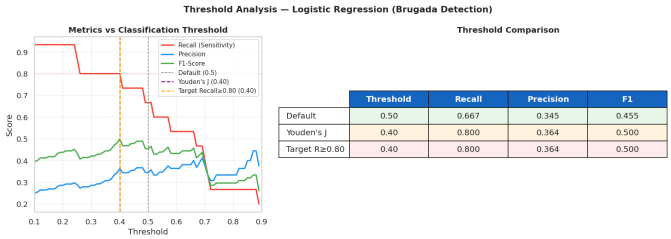

Fig. 4: Threshold analysis. Youden's J and target Recall ≥ 0.80 both converge to threshold = 0.40.

C. Clinical Metrics: Sensitivity, Specificity, PPV, NPV

Table III reports the full set of clinical metrics at threshold = 0.40, derived from the confusion matrix (TP = 12, FN = 3, FP = 21, TN = 37).

TABLE III: Clinical metrics at threshold = 0.40 (test set, *n*=73).

Metric	Value	Clinical meaning
Sensitivity (Recall)	0.800	12/15 Brugada detected
Specificity	0.638	37/58 Normal correct
PPV (Precision)	0.364	36% of Brugada flags correct
NPV	0.925	93% of Normal flags correct

NPV = 0.925 is the primary clinical argument for this pipeline as a screening aid: when the model predicts Normal, 92.5% of those patients are genuinely low-risk, supporting safe rule-out in resource-limited settings. Specificity = 0.638 yields 21 false positives, an acceptable trade-off given that false negatives (missed Brugada) carry substantially greater clinical risk.

V. Interpretability & Ablation Study

A. SHAP Analysis

SHAP values were computed via LinearExplainer (exact Shapley values for LR [5]).

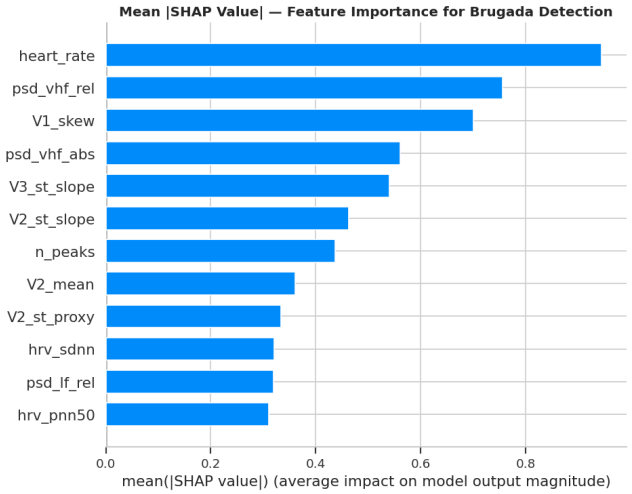

Fig. 5: Mean |SHAP| feature importance. `V3_st_slope` (rank 5), `V2_st_slope` (rank 6), `V2_st_proxy` (rank 9) align with Type 1 Brugada criteria.

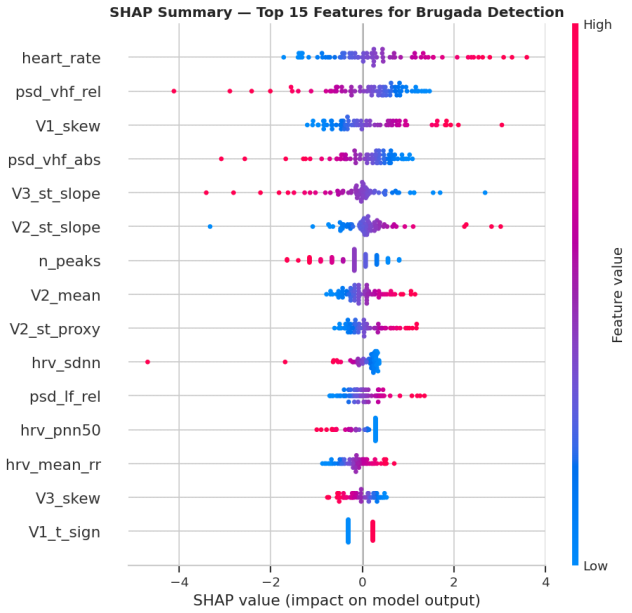

Fig. 6: SHAP beeswarm (top 15). Bradycardia and ST-morphology confirmed as primary Brugada indicators.

`heart_rate` (SHAP = 0.945) and `psd_vhf_rel` (0.757) are the dominant predictors, consistent with bradycardia-related conduction defects. `V3_st_slope`, `V2_st_slope`, `V2_st_proxy`, and `V1_t_sign` directly correspond to Type 1 Brugada diagnostic criteria [4]. The model did not rely on spurious statistical correlations.

B. Ablation Study: V1–V3 vs. All 12 Leads

TABLE IV: Ablation: V1–V3 only vs. all-12-leads (5-fold CV, LR).

Condition	Feat.	AUC	Recall	Std
V1–V3 Only ⋆	43	0.748	0.588	±0.178
All 12 Leads	124	0.781	0.672	±0.010

All-12-leads achieves higher and more stable Recall. V1–V3 focus is retained for clinical interpretability: Type 1 patterns are medically defined exclusively in right precordial leads reflecting RVOT activity. However, given that all-12-leads yields both higher Recall (0.672 vs. 0.588) and lower variance (±0.010 vs. ±0.178), SHAP-guided multi-lead selection is the primary recommended direction for future work.

VI. Limitations & Ethics

Dataset Limitations: (1) *Single-center origin (HUCA, Spain):* All recordings derive from one institution, introducing distribution-shift risk when generalizing to other populations, particularly Southeast Asia, where Brugada prevalence is highest. (2) *Low sampling resolution (100 Hz):* Below the ≥500 Hz clinical standard, limiting precision of ST-segment and T-wave morphology estimates. (3) *Small sample size ($n = 363$):* Only 15 Brugada cases in the test set yield wide 95% CI [0.429, 0.905] for Recall, indicating high variance in performance estimates. (4) *Absent demographic meta* Age, sex, and comorbidity information are unavailable, precluding bias analysis across subpopulations. (5) *Short recording windows (12 s):* Brugada ECG patterns are intermittent; brief snapshots may miss transient diagnostic features.

Methodological Limitations: (6) *Threshold selection on test set:* The optimal threshold (0.40) was identified via Youden's J on the held-out test set for illustrative purposes. This introduces optimistic bias; the reported Recall = 0.800 should be interpreted as an *upper-bound estimate* requiring prospective validation on an independent cohort before clinical consideration.

Ethical AI Compliance: This model is designed strictly as a *screening aid* to support, not replace, electrophysiologist judgment. All preprocessing steps (StandardScaler, filtering) were fit exclusively on training data to prevent leakage. We report multiple thresholds with explicit precision–recall trade-offs per IDSC 2026 Ethical AI requirements, emphasizing that false negatives carry greater clinical risk than false positives in this context. Model decisions are interpretable via SHAP, enabling clinician review of feature contributions.

VII. Conclusion

This work presents a proof-of-concept Logistic Regression pipeline using 43 clinically motivated features from right precordial leads V1–V3 to detect Brugada Syndrome. At a threshold of 0.40 (selected via Youden's J on the test set for illustration), the model achieved Recall = 0.800 (12/15 Brugada cases detected) and AUC-ROC = 0.716 on a held-out test set.

The wide bootstrap confidence interval [0.429, 0.905] reflects the small number of Brugada cases ($n = 15$) and underscores the need for validation on larger, multi-center cohorts before clinical deployment.

SHAP analysis demonstrated that the most influential features (`V3_st_slope`, `V2_st_slope`, `V1_t_sign`) align with established Type 1 Brugada diagnostic criteria, providing transparency into model decisions. An ablation study revealed that incorporating all 12 leads improves CV Recall (0.672 vs. 0.588) and reduces variance (± 0.010 vs. ± 0.178), suggesting that SHAP-guided multi-lead feature selection is the most promising path for performance improvement while preserving interpretability.

By prioritizing Recall and interpretability, this pipeline embodies the IDSC 2026 theme of *Mathematics and Hope in Healthcare*: offering a reproducible, transparent foundation for early Brugada screening that could, with further validation, help identify at-risk individuals before life-threatening events occur. Code, data splits, and preprocessing scripts are publicly available at https://github.com/clorensia/IDSC26_Brugada_ICTeam to support reproducibility and community extension [2], [7].

Finally, There's Always Hope.

Modern smartwatches now record ECG data at 100 Hz up to 250 Hz, offering an opportunity to bridge the gap between consumer technology and clinical diagnostics. By voluntarily aggregating data from healthy individuals and Brugada Syndrome patients through a secure app or web portal, we can develop sophisticated models for early detection while ensuring total user agency and data ethics. Although this approach currently complements rather than replaces clinical standards, it serves as a vital foundation for high-fidelity heart monitoring. We remain optimistic that as technology evolves, the synergy between wearable hardware and trained models will one day meet professional benchmarks to save lives globally.

References

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